

Danmaku Avatar: Enabling Co-viewing Experiences in Virtual Reality via Danmaku

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ABSTRACT

With the evolution of social interaction needs, danmaku comments on 2D screens have emerged in Asia, offering users a virtual co-viewing experience. With advancements in virtual reality (VR) technology, there is an increasing demand for similar co-viewing experiences in VR settings. In this study, we designed the Danmaku Avatar System to explore the adaptation of danmaku into VR environments to enhance shared viewing experiences. Comparative experiments showed that danmaku avatars substantially improved social presence and increased the desire for communication. These findings lay the groundwork for further refining danmaku representations in VR, advancing virtual social interactions and immersive viewing experiences.

Index Terms: Virtual Reality, social interaction, co-viewing, danmaku

1 INTRODUCTION

With the evolution of social interaction needs, danmaku video platforms, like Bilibili¹ and Niconico², have emerged and gained widespread popularity in Asia, offering users a virtual co-viewing experience when watching video content alone [3]. Danmaku refers to comments left by previous viewers at specific moments in a video, which are overlaid on the screen as new viewers watch the corresponding scenes (see Figure 1(a)). These comments appear in real-time, allowing viewers to share the perspectives and reactions of others who have watched the content before them.

Meanwhile, virtual reality (VR) technology is reshaping how people engage with media, with an increasing number of individuals choosing to watch videos and other content alone in VR environments [7]. While this mode of viewing offers a more immersive experience, marked by deep sensory engagement and heightened presence, the solitary nature of these experiences can occasionally lead to feelings of isolation [5]. This raises a crucial question: *Can danmaku, traditionally used as a form of virtual co-viewership on 2D screens, be effectively adapted to foster a similar sense of shared experience in VR?* This gap in research underscores important considerations for the future of social interactions in virtual spaces.

Several studies have explored the display of danmaku comments on 360° VR video screen [2] and the creation of virtual classmates using danmaku from online study videos [4]. However, these studies have not focused on harnessing the core value of shared viewing that danmaku provides, nor have they fully utilized the VR space's potential for simulation to transform danmaku into an immersive VR environment. In this study, we explore three key research questions: **RQ1:** What challenges do users encounter when co-viewing with danmaku on 2D screens, and what are their expectations for

danmaku co-viewing in VR environments? **RQ2:** Does the integration of danmaku-based virtual co-viewing in VR enhance the overall user experience? **RQ3:** How does the introduction of danmaku in VR influence viewer behavior?

2 RESEARCH DESIGN AND EVALUATION



Figure 1: The design process of the Danmaku Avatar System involves four key stages: (a) raw danmaku data; (b) processing and clustering the comments to map them onto avatars, which enable real-time interactions supported by LLMs and non-verbal behavior rules; (c) danmaku avatar choose panel; (d) experiencing video watching alongside danmaku avatars within a VR environment.

To address **RQ1**, we conducted an online survey with approval from the Institutional Review Board (IRB) to identify common challenges users face when co-viewing content with danmaku on 2D screens, as well as their expectations for watching video content in VR with danmaku. We received 132 valid responses. Our findings revealed that respondents view danmaku as a form of virtual co-viewer rather than just text comments. However, the text-only format and delayed responses diminished the sense of social presence. Respondents expressed a strong preference for a more immersive co-viewing experience in VR, particularly one with real-time interactive capabilities.

Based on the results of our user survey, we invited 13 students (7 males and 6 females) from a local research university, all with experience in VR system design and familiarity with danmaku culture, to participate in discussions about the system design. Following these discussions, we voted for the virtual agent with voice design from the five best candidates (Figure 2). Thus, we designed and implemented a Danmaku Avatar System in a virtual living room environment using Unity (Figure 1). The danmaku avatars were equipped with both verbal and non-verbal behaviors derived from danmaku comments. Additionally, we developed dialogue agents powered by a large language model (LLM) to provide real-time interactive support for the avatars [6]. To offer a more personalized viewing experience, we clustered the danmaku comments based on features such as emotional tone, content, and temporal attributes (Figure 1 (b)). This clustering allowed comments with similar characteristics to be represented by the same danmaku avatar, symbolizing a group of viewers with shared interests. Consequently, users could choose one or more danmaku avatars to accompany

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¹<https://www.bilibili.com/>

²<https://www.nicovideo.jp/>

them while watching a video, based on the avatars' focus of interest, emotional traits, and visual design (Figure 1 (c)). The danmaku avatar actively voiced danmaku comments and displayed corresponding gestures as the video progressed. Users could interact with the avatars, prompting them to respond based on both the ongoing plot and their previous comments, thereby enhancing the co-viewing experience (Figure 1 (d)).



Figure 2: Five design candidates following design discussions.

We then recruited 36 participants (20 females and 16 males) interested in the research through word-of-mouth and local university chat groups. To address **RQ2** and **RQ3**, the participants were randomly assigned into two groups of 18: one group watched the video alone with text format danmaku overlapping on the screen in a VR environment (*VR-A* group), the other group watched the video with danmaku avatars (*VR-DA* group). To compare the engagement levels of the two groups, we followed the methodology used by [8], utilizing electroencephalography (EEG) signals, specifically from the α (7-13 Hz), β (13-30 Hz), and θ (4-7 Hz) frequency bands. Engagement (E) was calculated using the formula $E = \beta / (\alpha + \theta)$ to track changes in engagement during the video viewing. During the experiment, participants were equipped with the *HTC VIVE headset* and *BrainLink EEG device* and underwent a 5-minute session for environmental adaptation and device operation practice. Following this, each group watched a 20-minute video. After the experiment, participants completed a questionnaire on social presence, using a 5-point Likert scale, and participated in a semi-structured interview. Social presence was assessed using three subscales from the social presence questionnaire (SPQ) [1]: Co-presence (CP), Perceived Message Understanding (PMU), and Perceived Affective Understanding (PAU).

3 RESULT

We employed both qualitative and quantitative methods to assess the effectiveness of danmaku avatars, focusing on engagement, social presence, and insights gathered from semi-structured interviews. The mean engagement score for the *VR-DA* group was 0.253, compared to 0.228 for the *VR-A* group, indicating a higher average for the *VR-DA* group. However, a t-test revealed a p-value of 0.502, suggesting no statistically significant difference between the two groups. In terms of social presence (Figure 3), the *VR-DA* group scored higher on all three subscales of the SPQ compared to the *VR-A* group: $VR-DA_{CP} = 4.533$, $VR-DA_{PMU} = 4.25$, $VR-DA_{PAU} = 0.373$; $VR-A_{CP} = 3.6$, $VR-A_{PMU} = 3.6$, $VR-A_{PAU} = 0.267$. One-way analysis of variance (ANOVA) showed that the *VR-DA* group significantly outperformed the *VR-A* group in the CP and PAU subscales ($p_{CP} < 0.05$, $p_{PAU} < 0.05$), suggesting that danmaku avatars enhance social presence by promoting a sense of co-presence and improving emotional connection among participants. Based on the interviews, we found that the higher standard deviation in PMU might be attributed to some participants' discomfort with using voice for danmaku. They believed that certain humor in danmaku could only be effectively conveyed through text. Moreover, participants reported feeling pleasure when the danmaku avatar expressed views similar to their

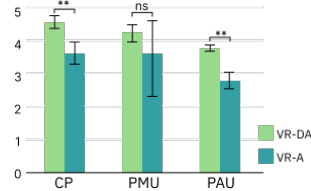


Figure 3: SPQ result compare. Error bars indicate standard deviations. (ns: $p > .1$, *: $p < .05$, **: $p < .01$)

own. All participants interacted with the danmaku avatar at least three times, with nearly all participants stating that the presence of the danmaku avatar increased their motivation to engage in communication.

4 CONCLUSION, LIMITATION, AND FUTURE WORK

The primary aim of this study is to explore the transformation of danmaku—a text-based co-viewing experience on 2D screens—into a VR environment, enabling a co-viewing experience in VR and examining the resulting changes in user viewing behavior. Our findings suggest that while danmaku avatars did not significantly impact user engagement, they did enhance social presence and the desire for social interaction. Consequently, further research is required to refine danmaku avatars for VR environments with the aim of improving user engagement and the viewing experience.

Although this study provides an initial approach to transforming danmaku into danmaku avatars and offers preliminary insights into its impact on the user viewing experience, it has several limitations that warrant further exploration. Currently, danmaku is clustered and mapped onto virtual human avatars, but future research could investigate more innovative representations of danmaku in VR environments to create new interactive and visual experiences. We hypothesize that offering users novel viewing scenarios and co-viewer representations could substantially enhance their immersive viewing experience.

In summary, this work contributes valuable insights into the study of viewing experiences in VR, particularly in the fields of human-computer interaction and user experience enhancement within virtual environments. It demonstrates how technology can be integrated with social behaviors to redefine how people co-view and interact in digital spaces, thus transforming the social aspects of content consumption. The exploration of the intersection between technology and social behavior could open up new avenues for applications in virtual social interactions and virtual reality.

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